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Estimates of relativistic contributions to molecular properties

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A simple method for the estimation of the relativistic contribution to atomic and molecular properties is proposed. The method assumes that the dominant portion of relativistic contribution to different properties is accounted for by the Cowan–Griffin operator. In contrast to recently investigated variational relativistic and quasirelativistic approaches, the present method is based virtually on the triple-perturbation theory approach and can be easily executed in the framework of the finite-field perturbation schemes. The method proposed in this paper is applied to the evaluation of the relativistic contribution to electric properties of hydrogen halides. This contribution turns out to be completely negligible in the case of HCl. For the HI molecule, the relativistic correction to its dipole moment becomes almost as important as the electron correlation effects. A similar result is also obtained in the case of the dipole moment of AgH. Different possible applications of the present method are discussed. The major advantage of the proposed perturbation approach is a very simple computational structure which permits the calculation of relativistic corrections from any available nonrelativistic wave function.

I. INTRODUCTION

The importance of including the relativistic effects in the electronic structure calculations for systems involving heavy atoms is already well recognized.[1-3] There is also a growing interest in different methods for the estimation of the magnitude of the relativistic contribution to different molecular properties.[1-4] So far, attention has been mostly focused on relativistic contributions to spectroscopic parameters derived from the calculated potential energy curves.[3-6] Among other properties the relativistic effects on molecular dipole moments have been recently studied by several authors.[3(b),4,7-9]

The study of relativistic effects can be carried out at various levels of approximation for both the so-called relativistic many-electron Hamiltonian[1-4] and the many-electron wave function. Within the one-electron model remarkably good results have been obtained from quasirelativistic approximations[10-12] to the Dirac–Fock Hamiltonian.[1,3,4] It has been argued that only the two leading terms of the relativistic origin, i.e., the mass-velocity and the one-electron Darwin contributions,[1] can be retained in the relativistic Hamiltonian.[10,13] The resulting quasirelativistic Hamiltonian of Cowan and Griffin[10] has been shown to perform reasonably well.[5,13,14] In spite of the neglect of spin–orbit interactions and several other terms it provides a very attractive tool for atomic and molecular calculations. Of particular importance is the explicit perturbation form of the Cowan–Griffin Hamiltonian[10,13] with the usual nonrelativistic term considered as the zeroth-order operator.

As long as the spin–orbit terms are negligible in comparison with those retained in the Cowan–Griffin operator, the perturbation treatment of the presumably dominant relativistic contribution can also be carried out at the correlated level of approximation.[5] Such an approximation might be sufficient for estimating the magnitude of relativistic contributions which would result from more elaborate methods.[3,4,6,14] Additionally, the explicit perturbation approach can also be employed for the investigation of the electron correlation contribution.[15] In the case of nondegenerate reference functions the many-body perturbation theory (MBPT) approach[15,16] appears to be particularly suitable for the evaluation of the electron correlation contributions to energies and properties of many-electron systems.

The notion of atomic and molecular properties follows from the consideration of a many-electron system subject to some (external) perturbation.[17] Thus, assuming that the one-electron approximation for the nonrelativistic isolated system offers a reasonable starting point, one can develop a perturbation theory which will handle simultaneously three perturbations responsible for the origin of the given property and the electron correlation and relativistic contributions to it.

A natural reference state for the triple-perturbation theory of relativistically and correlation-corrected properties is given by the single-determinant Hartree–Fock (HF) approximation.[15,17] This assumption leads to a particular form of the perturbation expansion[17,18] which permits a unique separation between the pure and the so-called apparent correlation contributions.[17] The latter arise from the modification of the HF potential in the presence of other perturbations. Such a perturbation theory, which reflects all features of the HF reference state, will be developed in Sec. II.

In Sec. III, the triple-perturbation theory of relativistic and correlation contributions to properties of many electron systems is applied for the investigation of dipole moments and polarizabilities in the series of hydrogen halides. The Cowan–Griffin operator[10] is used to represent the relativis-

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tic effects in those systems. The spin–orbit interaction terms are neglected as supported by the recent results of Chapman *et al.*[7(a)] The importance of the spin–orbit interaction contribution to molecular dipole moments increases, however, with the nuclear charge[7(b)] and its neglect imposes some limitations on applications of the present method.

The possibility of perturbationally including the relativistic effects within the methods which go beyond the single-determinant HF model is investigated at the level of the multiconfiguration SCF (self-consistent field) approximation. More specifically, the relativistic corrections to different properties of the AgH molecule are calculated with density matrices obtained within the complete active space (CAS) SCF method.[19]

The numerical results presented in Sec. III as well as the advantages and limitations of the triple-perturbation approach to the simultaneous investigation of the electron correlation and relativistic effects on atomic and molecular properties are discussed in Sec. IV. It is stressed that the method developed in this paper is not meant to give accurate values of the relativistic contribution to properties. The method is designed to produce estimates of the corresponding corrections which seem to be badly needed to resolve certain discrepancies between the current theoretical and experimental data.[8,20,21] The conclusions of this paper are presented in Sec. V.

II. THEORY

Both the external field and relativistic effects are considered to be a (small) perturbation to the nonrelativistic Born–Oppenheimer Hamiltonian H^0. Let H^{10} be the perturbation operator leading to the molecular property of interest. In the nonrelativistic case the perturbed Hamiltonian $H_{\rm NR}$,

$$H_{\rm NR} = H^0 + \lambda H^{10}, \tag{1}$$

where λ is the ordering parameter, has a λ-dependent energy spectrum. For the state of interest, whose eigenenergy is $E_{\rm NR}(\lambda)$, the k th order property related to H^{10}, say $Q_{\rm NR}^{(k)}$ is defined as a quantity proportional[18] to the k th derivative of $E_{\rm NR}(\lambda)$ with respect to λ, i.e.,

$$Q_{\rm NR}^{(k)} = Q^{(k,0)} \sim \left(\frac{\partial^k E_{\rm NR}(\lambda)}{\partial \lambda^k}\right)_{\lambda=0}. \tag{2}$$

Following the same method, the simultaneous perturbation treatment of the relativistic contribution gives the approximate relativistic Hamiltonian H_R,

$$H_R = H_{\rm NR} + \mu H^{01}, \tag{3}$$

where H^{01} represents some conveniently chosen perturbation operator which is expected to account for the dominant portion of the relativistic effects on the spectrum of $H_{\rm NR}$.[1,5,6,10–13] The parameter μ orders the relativistic perturbation series and the appropriate counterpart of Eq. (2) becomes

$$Q^{(k)} = Q^{(k)}(\mu)|_{\mu=1} = \sum_m Q^{(k,m)}, \tag{4}$$

where

$$Q^{(k,m)} \sim \left(\frac{\partial^{k+m} E_R(\lambda,\mu)}{\partial \lambda^k \partial \mu^m}\right)_{\lambda=0,\mu=0}, \tag{5}$$

and $E_R(\lambda,\mu)$ is the eigenenergy for the approximate relativistic perturbed Hamiltonian (6). Thus, the k th order property $Q^{(k)}$ is expressed in terms of the nonrelativistic value $Q_{\rm NR}^{(k)} = Q^{(k,0)}$ and a series of relativistic corrections $Q^{(k,m)}$, $m \geqslant 1$. For the given H^{01} Eq. (5) is the exact definition of the mth order relativistic correction to the k th order property. This definition in terms of the total energy derivatives is independent of the quality of the approximate wave function used to calculate $E_R(\lambda,\mu)$.[17,18,22,23]

The initial step in derivations of $Q^{(k,m)}$ requires assuming that the zeroth-order eigenvalue problem for H^0 is exactly solved. Actually, one knows only some approximate solutions of the unperturbed nonrelativistic Schrödinger equation,

$$H^0\Psi^0 = E\Psi^0, \tag{6}$$

and the perturbation treatment of

$$H_R(\lambda,\mu)\Psi_R(\lambda,\mu) = E_R(\lambda,\mu)\Psi_R(\lambda,\mu) \tag{7}$$

must be adapted to those circumstances.[17,18]

A convenient approximation to Ψ^0 of Eq. (6) is given by the HF method and splits the unperturbed Hamiltonian into its HF $H_{\rm HF}^0$ and correlation perturbation $H_{\rm corr}^0$, components[15–18]

$$H^0 = H_{\rm HF}^0 + \sigma H_{\rm corr}^0. \tag{8}$$

On using the HF approximation for Ψ_R of Eq. (7) one obtains, in analogy to Eq. (8), the following partition of the perturbed Hamiltonian (3):

$$H_R = H_{R,\rm HF} + \sigma H_{R,\rm corr}, \tag{9}$$

where both components of the right-hand side of Eq. (9) can be considered as λ,μ dependent,

$$H_{R,\rm HF} = H_{R,\rm HF}(\lambda,\mu), \tag{10}$$

$$H_{R,\rm corr} = H_{R,\rm corr}(\lambda,\mu), \tag{11}$$

and σ is again a formal ordering parameter for the electron correlation perturbation series.[17,18]

It is worth while to remark that the partition of the total Hamiltonian, which is formally analogous to that in Eq. (9), can be also employed in a nonperturbative treatment of relativistic effects.[24,25] Such an approach leads to the relativistic theory of the electron correlation contribution to (relativistic) electromagnetic properties of atoms and molecules. The HF Hamiltonian of Eq. (8) becomes than the Dirac–Fock Hamiltonian of the relativistic single-particle model.[26,27] Including the retardation terms[2] is also possible.[24,25]

In the present paper our considerations will be limited to a rather standard perturbation approach based on partition (9) and a simple form of the relativistic perturbation operator.[5,10] Then, partition (9) of the total quasirelativistic energy E_R of Eq. (7) can be written as

$$E_R = E_R(\lambda,\mu,\sigma)|_{\lambda=1,\mu=1,\sigma=1}$$
$$= \sum_{k,m,n} E^{(k,m,n)} = \sum_{k,m,n} c_k Q^{(k,m,n)}, \tag{12}$$

where c_k are the appropriate proportionality constants between the (m,n)th order relativistic-correlation corrections to the kth order property $Q^{(k)}$ and the perturbed energies $E^{(k,m,n)}$.

The simplified choice for the relativistic perturbation in the form of the Cowan–Griffin operator suggests that a consistent perturbation expansion should not be carried out beyond the first-order ($m \leqslant 1$) in H^{01}. Thus, the kth order property corrected through the first order in H^{01} will have the following expansion:

$$Q^{(k)} = \sum_n [Q^{(k,0,n)} + Q^{(k,1,n)}], \tag{13}$$

where the first term combines all pure correlation contributions which are recovered at the level of the nonrelativistic treatment.[17,18] In the case of the nonrelativistic SCFHF approach the perturbation treatment based on partition (9) is known[17] to have

$$Q^{(k,0,1)} \equiv 0, \quad k > 0. \tag{14}$$

The second term on the right-hand side of Eq. (13) can be partitioned into a pure first-order relativistic correction $Q^{(k,1,0)}$, and the cross terms $Q^{(k,1,n)}$, $n > 1$, which represent the mutual interference of the correlation and relativistic perturbations. The latter terms are expected to be of secondary importance in comparison with $Q^{(k,0,n)}$ and $Q^{(k,1,0)}$. Also their numerical evaluation by standard methods of computational quantum chemistry may pose certain formal problems.

The evaluation of pure correlation components of $Q^{(k)}$ is conveniently carried by using the MBPT methods.[16–18] Whenever the SCFHF reference is a judicious choice for the perturbation treatment, the pure correlation perturbation series in Eq. (13) can be usually limited to a few leading terms;[28–30] the second-order approximation

$$Q^{(k,0)}(2) = Q^{(k,0,0)} + Q^{(k,0,2)}, \quad k > 0, \tag{15}$$

being by no means a poor one.[28–30] Under the present assumptions concerning the reference state, the fourth-order treatment of the electron correlation

$$\begin{aligned} Q^{(k,0)}(4) = {} & Q^{(k,0,0)} + Q^{(k,0,2)} \\ & + Q^{(k,0,3)} + Q^{(k,0,4)}, \quad k > 0 \end{aligned} \tag{16}$$

usually gives very satisfactory results for nonrelativistic atomic and molecular properties.

It is of particular importance that for a wide class of properties the pure correlation contribution is primarily determined by the valence correlation effects, while the relativistic correction from the Cowan–Griffin operator[10]

$$H^{01} = H^{01}_{CG} = -\sum_j \frac{1}{8c^2} \nabla_j^4 + \sum_A \sum_j \frac{\pi}{2c^2} Z_A \delta(r_{Aj}) \tag{17}$$

comes mostly from the regions close to the nuclei A of the given system. The approximate relativistic perturbation operator (17) is expressed in atomic units and $c = 137.036$ is the velocity of light. The summation over j runs over all electrons in the system whose nuclei have the charge Z_A.

Among the properties which belong to the abovementioned class, one finds the electric multipole moments and polarizabilities. The disjoint character of the valence and core contributions makes the approximation

$$Q^{(k)}(n) = \sum_{p=0}^{n} Q^{(k,0,p)} + Q^{(k,1,0)} \tag{18}$$

fairly justified. Thus, the correlation and relativistic contributions can be calculated separately with the nonrelativistic SCFHF reference function. In Sec. III B, this approach is employed for the investigation of dipole moments and polarizabilities in the hydrogen halide series. The MBPT methods are used to evaluate the pertinent electron correlation contributions.

One should mention, however, that the present perturbation theory of the correlation-relativistic contributions to $Q^{(k)}$ is by no means restricted to the use of the MBPT expansion. The theory can be easily rewritten for any form of the reference function, including those which are supposed to globally account for the electron correlation contribution and are based on the configuration interaction (CI) ansatz. The methods used to evaluate $Q^{(k)}$ depend then on whether or not the given wave function satisfies the Hellman–Feynman theorem.[22,23]

The only legitimate way to define $Q^{(k)}$, which is simultaneously compatible with all degrees of freedom involved in an arbitrary wave function $\Psi_R(\lambda,\mu)$, is to express it in terms of the energy derivative. If $E_R(\lambda,\mu)$ is the approximate expectation value of energy

$$E_R(\lambda,\mu) = \frac{\langle \Psi_R(\lambda,\mu) | H_R(\lambda,\mu) | \Psi_R(\lambda,\mu) \rangle}{\langle \Psi_R(\lambda,\mu) | \Psi_R(\lambda,\mu) \rangle}, \tag{19}$$

which corresponds to the given $\Psi_R(\lambda,\mu)$, then $Q^{(k)}$ should be determined from Eqs. (4) and (5).

The case when $\Psi_R(\lambda,\mu)$ satisfies the Hellmann–Feynman theorem deserves particular attention.[17,18] Then,

$$Q^{(1)} \sim E^{(1)} = \langle \Psi_R(0,\mu) | H^{10} | \Psi_R(0,\mu) \rangle |_{\mu=1} \tag{20}$$

and

$$Q^{(k)} \sim \left[\frac{\partial^{k-1}}{\partial \lambda^{k-1}} \langle \Psi_R(0,\mu) | H^{10} | \Psi_R(0,\mu) \rangle \right]_{\lambda=0,\mu=1}. \tag{21}$$

In a similar way one obtains

$$\begin{aligned} \sum_k c_k Q^{(k,1)} &= \left[\frac{\partial}{\partial \mu} E_R(\lambda,\mu) \right]_{\mu=0,\lambda=1} \\ &= \langle \Psi_{\rm NR}(\lambda) | H^{01} | \Psi_{\rm NR}(\lambda) \rangle |_{\lambda=1}, \end{aligned} \tag{22}$$

where $\Psi_{\rm NR}(\lambda)$ corresponds to the solution of the nonrelativistic Schrödinger equation (6) with the perturbed Hamiltonian (1) and $\Psi_{\rm NR}(\lambda) = \Psi_{\rm NR}(\lambda,\mu=0)$. The important result of the Hellmann–Feynman theorem, which follows from Eq. (22), is

$$c_k Q^{(k,1)} = \left(\frac{\partial^k}{\partial \lambda^k} \langle \Psi_{\rm NR}(\lambda) | H^{01} | \Psi_{\rm NR}(\lambda) \rangle \right)_{\lambda=0}. \tag{23}$$

This means that the first-order relativistic correction to the kth order property resulting from H^{10} can be evaluated as the appropriate derivative of the Hellmann–Feynman result (22) calculated with λ-dependent, nonrelativistic wave function $\Psi_{\rm NR}(\lambda)$. Since according to Eq. (17) the relativistic perturbation operator employed in this paper is a sum of one-electron operators, the evaluation of Eq. (22) involves solely the one-particle density matrix of $\Psi_{NR}(\lambda)$.

The very simple result (23) requires that the nonrelativistic function $\Psi_{NR}(\lambda,\mu)$ satisfies the Hellmann–Feynman theorem. This condition is met, for instance, by multireference SCF wave functions.[19,28] However, a limited (incomplete) CI wave function[22,23] built from some arbitrary set of single-particle functions will not, in general, satisfy Eq. (20). In spite of violation of the Hellmann–Feynman theorem by such functions, the evaluation of properties is frequently based on Eqs. (20)–(23). The expectation value definition has been employed also by Chapman *et al.*[7] So far, its validity in relativistic limited-CI calculations has not been examined.

As an illustration of different methods which can be used for the evaluation of the first-order relativistic contribution to properties of many-electron systems we have employed Eq. (23) in the case of the CASSCF approximation for $\Psi_{NR}(\lambda)$.[19] The method used to obtain $\Psi_{NR}(\lambda) = \Psi_{CASSCF}(\lambda)$ satisfies the Hellmann–Feynman theorem[19] and the use of Eq. (23) is perfectly justified. The results for the first- and second-order properties calculated by this method will be shown in Sec. III C.

The important difference between the truncated MBPT series result (18) and that of Eq. (23) follows from the fact that $\Psi_{NR}(\lambda)$ in Eq. (23) may already correspond to certain correlated level of approximation. Then, derivative (23) comprises automatically some of the correlation-relativistic terms which have been neglected in the MBPT series (18). Obviously, those terms enter the complete MBPT correlation perturbation series for $Q^{(k,1)}$. However, their explicit evaluation may lead to certain computational problems.

Finally, let us mention that the derivatives entering Eqs. (12) and (23) can be evaluated either analytically or numerically. For low order properties $Q^{(k)}$, the numerical approach offers several computational advantages and is employed in the present paper. Moreover, the relativistic perturbation can also be considered in a purely numerical (finite-field) manner. For the simplified Cowan–Griffin form of H^{01} this appears to be a convenient method leading to correlation-relativistic corrections $Q^{(k,l,n)}$, $n>1$ to $Q^{(k)}$.

III. CALCULATIONS

A. Computational details

All calculations reported in this paper have been carried out by using the newly developed MOLCAS system of quantum chemistry programs.[30] The computational methods employed in the present study are well described in different articles[16,19] and there is no need to repeat the corresponding details.

The calculation of various contributions which arise from the triple-perturbation series discussed in Sec. II, has been performed by using the finite-field perturbation methods for the external electric field perturbation.[32] The dipole moments and polarizabilities and the correlation and relativistic corrections to those properties are evaluated as the numerical derivatives of the appropriate electric field dependent energy expressions. Thus, the method employed in this study follows directly the definition of the corresponding terms in the correlation-relativistic series for the given property.

As an illustration of the method exposed in Sec. II, we have calculated the dipole moments and polarizabilities of a series of hydrogen halides: HCl through HI. A similar study at the level of the CASSCF approximation has been carried out for the AgH molecule.

The calculations for the hydrogen halide series have been carried out with large basis sets of Gaussian-type orbitals (GTO). The form of the Cowan–Griffin operator (17) used for the perturbation representation of the relativistic contribution requires an accurate and flexible basis set in vicinity of the heavy nucleus. On the other hand, the form of the electric field perturbation sets up a similar requirement upon the outer (valence) regions of the electron density distribution. To satisfy those two requirements, the halogen atom basis sets were left completely uncontracted in the s and p subsets. Moreover, the range of the tabulated orbital exponents for s- and p- type GTOs has been extended. The polarization functions have been chosen following the basis set polarization method.[33]

The initial (12.9) GTO basis set[34] for Cl has been augmented with one tight GTO (exponent $\alpha = 1\,559\,693.7$) s-type GTO and two diffuse s-type GTOs ($\alpha = 0.069\,659$ and $0.025\,077$). A similar extension has been carried out in the p subset ($\alpha = 3327.6$, $0.043\,75$, and $0.015\,31$); the orbital exponents being derived from considerations of the corresponding approximate even-tempered sequences. The basis set polarization method[33] gave a subset of seven d-type GTOs (orbital exponents in the range 16.6121 to 0.043 75 are the same as those in the p subset) and two f-type GTOs ($\alpha = 0.3580$ and 0.1250). The d subset has been a little contracted in the range of the three highest exponents; the contraction coefficients being determined from the electric field polarization of the $3p$ SCFAO. This has led to the [15.12.7.2/15.12.5.2] GTO/CGTO basis set for Cl.

In a similar way the [17.14.8.2/17.14.6.2] GTO/CGTO basis set has been derived for Br starting from the (14.11.5) GTO set of Huzinaga.[35] The added s ($\alpha = 7\,648\,346.$, 0.0593, and 0.025), p ($\alpha = 22\,026$, 0.0427, and 0.0132), and d ($\alpha = 0.4423$, 0.1378, and 0.0427) primitive GTOs have been chosen to increase the initial basis set flexibility in the regions important for the perturbation operators considered in this study.

The initial (21.17.12) primitive GTO basis set of iodine[36] appears to have sufficiently high exponents for s and p functions. Its extension has been carried out only in the valence region by adding two s ($\alpha = 0.050\,23$ and $0.022\,96$) and one p ($\alpha = 0.016\,75$) functions. The d subset has been augmented by three primitive d-type GTOs ($\alpha = 0.240\,41$, $0.109\,89$, and $0.044\,29$) and then partly contracted in the range of high orbital exponents. The addition of two f-type polarization functions ($\alpha = 0.109\,89$ and $0.044\,29$), selected according to basis set polarization scheme,[35] has finally led to the [23.18.15.2/23.18.9.2] GTO/CGTO basis set employed in calculations for HI.

The [11.7.2/7.4.2] GTO/CGTO basis set for hydrogen is the same as that employed in our previous calculations of electric properties of the LiH molecule.[35] Since the relativistic contribution of hydrogen is of negligible importance,[1,2,37] its GTO/CGTO basis set is selected mostly for a proper representation of the valence polarization effect[33] in external electric fields.

In the case of the AgH molecule, the main purpose of our study was to investigate the mutual dependence of the valence shell correlation and the relativistic effects on its dipole moment and polarizability. The GTO/CGTO studies of the AgH molecule have been carried out by several authors[5,6,38,39] at different levels of approximation. For the purpose of comparison with some earlier results[5,6] our calculations have been carried out with a rather small GTO/CGTO basis set employed previously by Martin[5] and Hess and Chandra.[6] The [15.10.8/15.10.6] basis set[5,6] for Ag and the [9.2/6.2] set[6] for H may not be sufficiently flexible to correctly account for the high polarity of AgH and its dependence on the electron correlation effects. On the other hand, the AgH basis set of Martin[5] should well enough represent the relativistic perturbation operator (17).

In calculations for AgH we have closely followed the computational details of earlier papers.[5,6] Hence, all six components of Cartesian d-type functions were retained in the basis set. In the case of the hydrogen halide molecules the s components of the d-type and p components of the f-type GTOs are removed from the the final molecular basis set. Moreover, in order to reduce the timing of the complete fourth-order valence MBPT calculations for HI several high-energy virtual molecular orbitals have been deleted. Under the orbital energy threshold of about 10^3 hartree, 14 σ- and 10 π-type molecular orbitals are deleted.

The MBPT calculations of the electron correlation contribution to energies and electric properties of hydrogen halides have been carried out through the complete fourth-order [MBPT(4)][16–18] only in the valence approximation. Also the approximate fourth-order approach [SDQ-MBPT(4)],[16–18] in which the contribution of triply substituted intermediate states is neglected, has been investigated. The study of the inner shell electron correlation contribution is limited to the second-order MBPT approach [MBPT(2)]. However, in most cases the MBPT(2) level of approximation gives the dominant portion of the electron correlation correction to the SCF energies and properties of many-electron systems. The all-electron MBPT(2) data can be combined with the fourth-order valence results leading to an estimate of the electron correlation contribution due to all electrons in the given molecule.

The study of the electron correlation effects in the AgH molecule has been carried out by using the CASSCF method.[19] Two sets of CASSCF calculations have been performed. First, we have computed the energy and electric properties of AgH with a CASSCF wave function built of two σ-type active orbitals for two valence electrons. In the second set of calculations the active orbital space was built of 4 σ, 2 π, and 2 δ molecular orbitals for 12 electrons.

B. Hydrogen halides

The calculations for HCl, HBr, and HI have been performed at different levels of approximation whose choice reflects the ideas leading to Eq. (18). The pure valence correlation effects have been investigated at the level of the fourth-order MBPT method. The importance of the electron correlation contribution of the halogen core electrons has been studied only at the level of the second-order MBPT approach. Though the higher orders of the MBPT expansion give a considerable contribution to the total core and core–valence correlation energies, they are rather unlikely to significantly influence the dipole moment and polarizability values.

The relativistic correction to energies and electric properties of hydrogen halides has been evaluated at the SCF level of approximation with the Cowan–Griffin perturbation operator (17). As already mentioned in Sec. II, the electric field and relativistic perturbations affect different regions of the electron density distribution, and thus the contribution from $Q^{(k,1,n)}$, $n \geqslant 2$, should be relatively small.

The main results of our calculations for HCl, HBr, and HI are presented in Tables I–III. In addition to calculations performed at the corresponding experimental equilibrium bond distances (R_e),[40] we have evaluated the energy and electric property data in the vicinity of R_e ($R_e \pm 0.05$ bohr). These results have led to the estimates of theoretical values of R_e, harmonic frequencies (ω_e), and vibrational corrections[41] to the dipole moment (μ). The bond distances and frequencies calculated in this paper are compared in Table IV with the experimental data. Our results for the dipole moment and polarizability of hydrogen halides combining the correlation, relativistic, and vibrational corrections to the nonrelativistic SCF values are presented in Table V. They will be analyzed and discussed in Sec. IV.

C. AgH

The AgH molecule represents a common test system for investigations of different approximations for relativistic many-electron Hamiltonians.[5,6,38] Its electronic structure is relatively simple and a qualitatively correct description of its $X\,^1\Sigma^+$ state can be achieved in terms of a two-configuration wave function built from the top occupied 9σ orbital and the corresponding σ-type correlating orbital. Within the CASSCF method, this corresponds to choosing a (2σ) active orbital space for two valence electrons of the system. A more extended CASSCF treatment may involve the two top occupied σ, one π and one δ orbitals and the same number of correlating orbitals in each symmetry. This $(4\sigma 2\pi 2\delta)$ active orbital space for 12 electrons leads to 57 168 configuration state function in the C_{2v} symmetry.

The principal advantage of the CASSCF method follows from the fact that the CASSCF function determined from the minimization of the energy functional (19) satisfies the Hellman–Feynman theorem.[19] Thus, the first-order relativistic correction to the CASSCF energy calculated from Eq. (22) combines the electron correlation and external electric field effects. As discussed in Sec. II, this leads to a convenient computational method to study the interplay between the electron correlation and relativistic contributions to molecular electric properties.

The present SCF and CASSCF results calculated at the internuclear distance $R = 3.30$ bohr are compiled in Table VI. Their discussion is given in the next section.

TABLE I. Energies and electric properties of HCl at $R_e = 2.409$ bohr.[a] All values in a.u.

Approximation[b]	Energy	Dipole moment	Dipole polarizability[c]
$Q^{(k,0,0)} + Q^{(k,0,1)}$[d]	−460.106 169	0.468 65	17.93
$Q^{(k,1,0)}$	−1.426 287	−0.004 82	0.02
$Q^{(k,0,2)}$ (all electrons)	−0.497 798	−0.030 83	0.59
(K shell frozen)	−0.443 399	−0.031 01	0.60
$Q^{(k,0,2)}$ (valence)	−0.196 389	−0.030 65	0.61
$Q^{(k,0,3)}$ (valence)	−0.019 652	−0.003 37	−0.14
$Q^{(k,0,4)}$ (valence)	−0.006 915	−0.008 63	0.16
(SDQ-valence)[e]	0.000 030	−0.003 68	−0.01
$Q^{(k)}(4)$, Eq. (18)[f]	−461.755 412	0.421 18	18.58
$Q^{(k)}(4)$, Eq. (18), est.[g]	−462.057 820	0.421 00	18.56
Reference results			
Ref. 42 Numerical SCF		0.463 8	
Ref. 43 SCF	−460.111 9	0.478 0	
Ref. 44 SCF[h]	−460.103 050	0.470 2	
CEPA[h,i]	−0.222 997	−0.036 6	
Ref. 45 SCF	−460.112 3	0.476 7	
SD-CI[j]	−0.437 0	−0.018 2	
MRSD-CI[j]	−0.445 5	−0.019 7	

[a] Reference 40.
[b] $Q^{(k,m,n)}$ is the mth order relativistic, nth order correlation contribution to the kth order property; $k = 0$ for the energy, $k = 1$ for the dipole moment, and $k = 2$ for the dipole polarizability. See Sec. II.
[c] Parallel component of the dipole polarizability tensor.
[d] SCF values.
[e] SDQ-MBPT (4) approximation for the fourth-order correlation contribution. See Sec. III A.
[f] Valence correlation contribution only.
[g] The second-order correlation contribution calculated for all electrons in the system while the third- and fourth-order results are those of the valence shell approximation.
[h] Bond distance $R = 2.409\,45$ bohr.
[i] Valence correlation contributions.
[j] All electrons correlated.

IV. DISCUSSION

The most representative data obtained in this study are the energy and electric property values computed for hydrogen halides. A convenient way of discussing the energy data involves the approximate values of R_e and ω_e listed in Table IV. The mutual comparison of those results computed in

TABLE II. Energies and electric properties of HBr at $R_e = 2.673$ bohr.[a] All values in a.u.

Approximation[b]	Energy	Dipole moment	Dipole polarizability[c]
$Q^{(k,0,0)} + Q^{(k,0,1)}$[d]	−2572.977 508	0.376 40	24.74
$Q^{(k,1,0)}$	−31.834 109	−0.017 91	0.00
$Q^{(k,0,2)}$ (all electrons)	−0.791 250	−0.035 18	+0.56
(K shell frozen)	−0.741 110	−0.035 23	+0.56
(K and L shells frozen)	−0.600 214	−0.035 03	+0.56
$Q^{(k,0,2)}$ (valence)	−0.173 948	−0.040 54	+0.65
$Q^{(k,0,3)}$ (valence)	−0.018 418	−0.004 74	−0.15
$Q^{(k,0,4)}$ (valence)	−0.006 550	−0.007 63	+0.13
(SDQ-valence)[e]	−0.003 236	−0.003 07	−0.04
$Q^{(k)}(4)$, Eq. (18)[f]	−2605.010 534	0.305 58	25.37
$Q^{(k)}(4)$, Eq. (18), est.[g]	−2605.627 836	0.310 94	25.28
Reference results			
Ref. 44 SCF	−2572.379 990	0.364 8	
CEPA[h]	−0.172 565	−0.042 5	
Ref. 46 SCF	−2572.980 807	0.374 4	24.34
SD-CI[h]	−0.180 320	−0.043 7	0.10
ACPF[h]	−0.191 139	−0.051 9	0.21
Ref. 9 MRSD-RCI[i]		0.322 7	
Ref. 48 SD-CI[j]		0.298 2	

[a–g] See the corresponding footnotes to Table I.
[h] Valence correlation contributions.
[i] Total dipole moment from relativistic MRSD-CI (MRSD-RCI) calculations with effective core pseudopotentials. Table III of Ref. 9.
[j] Total dipole moment from SD-CI calculations with semiempirical relativistic core pseudopotentials.

TABLE III. Energies and electric properties of HI at $R_e = 3.040$ bohr.[a] All values in a.u.

Approximation[b]	Energy	Dipole moment	Dipole polarizability[c]
$Q^{(k,0,0)} + Q^{(k,0,1)}$[d]	− 6918.566 102	0.260 12	37.13
$Q^{(k,1,0)}$	− 182.016 778	− 0.039 36	− 0.31
$Q^{(k,0,2)}$ (all electrons)	− 0.836 441	− 0.050 76	0.86
(K shell frozen)	− 0.785 588	− 0.050 78	0.86
(K, L shells frozen)	− 0.651 545	− 0.050 74	0.85
(K, L, M shells frozen)	− 0.365 466	− 0.050 87	0.86
$Q^{(k,0,2)}$ (valence)	− 0.136 562	− 0.055 62	0.96
$Q^{(k,0,3)}$ (valence)	− 0.016 446	− 0.004 85	− 0.14
$Q^{(k,0,4)}$ (valence)	− 0.005 540	− 0.008 37	+ 0.20
(SDQ-valence)[e]	− 0.001 081	− 0.002 13	− 0.04
$Q^{(k)}(4)$, Eq. (18)[f]	− 7100.741 428	0.151 92	37.84
$Q^{(k)}(4)$, Eq. (18), est.[g]	− 7101.441 307	0.156 78	37.74
Reference results			
Ref. 47 SCF	− 6912.598 848	0.240 4	
MCSCF[h]	− 0.036 145	− 0.031 9	
CEPA[h]	− 0.163 266	− 0.054 7	
Ref. 9 MRSD-RCI[i]		0.181 0	
Ref. 7(a) MRSD-RCI[j]		0.194 0	
Ref. 48 SD-CI[k]		0.182 0	

[a–g] See the corresponding footnotes to Table I.
[h–i] See the corresponding footnotes to Table II.
[j] Total dipole moment from relativistic MRSD-CI (MRSD-RCI) calculations with effective core pseudopotentials.
[k] Total dipole moment from SD-CI calculations with semiempirical relativistic core pseudopotentials.

different approximations shows the relative importance of the electron correlation and relativistic contributions to the shape of the potential energy curve. The pure electron correlation effect approximated by the fourth-order MBPT method and restricted to valence electrons increases the bond length and decreases the ω_e value in hydrogen halides. For R_e the contribution of the core correlation effects, which is estimated here from all-electron MBPT(2) results, has the opposite sign as compared to the valence MBPT correction. This effect rather rapidly increases with the nuclear charge of the halogen atom and is by no means negligible for HI. Both the valence and core correlation effects decrease the ω_e values.

The relativistic contribution, as estimated from our perturbation considerations, regularly increases with the heavy atom nuclear charge. Its effect can already be seen on the R_e and ω_e values for HBr and becomes definitely important for HI.

In spite of a number of different approximations involved in the calculation of both the electron correlation and relativistic contributions to the spectroscopic properties of hydrogen halides, it is rather tempting to compare our results with the experimental data. The results obtained for HCl show that the MBPT(4) method does not give a complete saturation of the electron correlation contribution to R_e and ω_e. This incompleteness of the finite-order treatment of the electron correlation effects seems to be the major source of small discrepancies between the present theoretical

TABLE IV. Estimated equilibrium bond distances (R_e in bohr) and harmonic frequencies (ω_e in cm^{-1}).

Method	HCl		HBr		HI	
	R_e	ω_e	R_e	ω_e	R_e	ω_e
SCF	2.394	3067	2.662	2758	3.046	2473
SCF + MBPT[a]	2.422	3013	2.691	2719	3.056	2428
SCF + MBPT + R[b]	2.422	3009	2.686	2693	3.039	2368
SCF + MBPT + R (est.)[c]	2.420	3003	2.681	2683	3.020	2354
Expt.[d]	2.4086	2991	2.6729	2648	3.0409	2309
Reference theoretical values	2.415[e]	2987[e]	2.680[e]	2644[e]	3.054[f]	2364[f]
			2.69[g]	2561[g]	3.14[h]	2305[h]

[a] Calculated from the MBPT(4) data in the valence approximation.
[b] Calculated from the valence MBPT(4) data and relativistic corrections; see Eq. (18).
[c] Estimated from the valence MBPT(4) data, the MBPT(2) core contributions and the relativistic correction.
[d] Reference 40.
[e] Reference 44, nonrelativistic valence CEPA results.
[f] Reference 47, nonrelativistic valence CEPA results.
[g] Reference 9, relativistic MRSD-CI calculations with effective core potentials.
[h] Reference 7(a), relativistic MRSD-CI calculations with effective core potentials.

TABLE V. Dipole moments μ and polarizabilities $\alpha_{\parallel}$ of hydrogen halides.[a] All values in a.u.

	μ_e	μ_{00}[b]	$\alpha_{\parallel,e}$
HCl ($R_e = 2.409$ bohr)			
SCF	0.469	0.472	17.93
SCF + MBPT[c]	0.426	0.428	18.55
SCF + MBPT + R[d]	0.421	0.423	18.57
SCF + MBPT + R(est.)[e]	0.421	0.423	18.56
Expt.[f]	0.4302	0.4362	
Ref. 44	0.435		
Ref. 50	0.438		18.405
HBr($R_e = 2.673$ bohr)			
SCF	0.376	0.378	24.74
SCF + MBPT[c]	0.323	0.324	25.37
SCF + MBPT + R[d]	0.306	0.307	25.37
SCF + MBPT + R(est.)[e]	0.311	0.312	25.28
Expt.[f]	0.3218	0.3252	
Ref. 44	0.323		
Ref. 46	0.3255		24.55
Ref. 9	0.323		
Ref. 48	0.2982		
HI($R_e = 3.040$ bohr)			
SCF	0.260	0.260	37.13
SCF + MBPT[c]	0.192	0.192	38.15
SCF + MBPT + R[d]	0.152	0.152	37.84
SCF + MBPT + R(est.)[e]	0.157	0.157	37.74
Expt.[f]	0.1760	0.1761	
Ref. 47	0.186	0.186	
Ref. 9	0.181		
Ref. 7(a)	0.194		
Ref. 48	0.182		

[a] Only the parallel component of the dipole polarizability tensor has been calculated.
[b] Estimated (Ref. 41) for the lowest vibrational level with vibrational corrections calculated from the dipole moment derivatives and experimental spectroscopic data from Ref. 40.
[c] Electron correlation contribution from the valence MBPT(4) approximation.
[d] Corrected for the valence correlation contribution at the level of the MBPT(4) approximation. The relativistic contribution according to Eq. (18).
[e] Including the estimated core correlation contribution.
[f] Reference 49.

results and the experimental data. Moreover, it should be recalled that the values of R_e and ω_e calculated in this paper follow from the simplest three-point approximation for the potential energy curve, and thus they may rather indicate the general trends for different corrections. In this respect, the dipole moment and polarizability data summarized in Table V seem to give a better account of the performance of the method used in this paper.

The dipole moments of hydrogen halides have been calculated by different authors and at different levels of approximation. Only some representative data are included in Tables I–III and V. Among them the recent relativistic multireference SD-CI values for HBr[9] and HI[7] seem to be of particular importance.

The electron correlation contribution to the dipole moment of all hydrogen halides studied in this paper appears to

TABLE VI. SCF and CASSCF results for the $X\,^1\Sigma^+$ state of AgH at $R = 3.30$ bohr. All values in a.u.

Method		Energy	Dipole moment[a]	Dipole polarizability[b]
SCF	Nonrelativistic	− 5197.530 081	− 2.0334	37.64
	Rel. correction[c]	− 108.644 588	0.1872	1.13
	Total	− 5306.174 668	− 1.8462	38.77
CASSCF	(2σ)			
	Total nonrelativistic	− 5197.549 409	− 1.7742	51.81
	Rel. correction[c]	− 108.645 129	+ 0.2011	− 0.76
	Total	− 5306.194 538	− 1.5731	51.05
CASSCF	($4\sigma 2\pi 2\delta$)			
	Total nonrelativistic	− 5197.639 202	− 1.7386	50.52
	Rel. correction[c]	− 108.647 458	+ 0.2026	− 0.52
	Total	− 5306.286 660	− 1.5360	50.00

[a] The negative sign corresponds to the polarity Ag^+H^-.
[b] The parallel component of the polarizability tensor.
[c] Relativistic corrections as estimated via the Hellmann–Feynman theorem, see Eq. (23).

be overestimated by the valence MBPT(4) contribution. This is a rather common feature of the MBPT(4) approach[16–18,51] and follows from usually overestimated contribution of triply substituted intermediate states.[16] The corresponding data can be read from Tables I–III as a difference between the complete MBPT(4) and approximate SDQ-MBPT(4) results. Since a considerable part of the fourth-order valence contribution to the dipole moment is accounted for by this difference, its value gives a measure of the uncertainties involved. If the SDQ-MBPT(4) values of Tables I–III were accepted to approximate the total valence correlation contribution, the final SCF + MBPT + R results of Table V would increase by about 0.005 a.u.

Another unknown in accurate calculations of molecular dipole moments is the contribution of the core and core–valence correlation effects. As indicated by the data of Tables I–III freezing the deep cores has practically no effect on the correlation contribution to μ. However, the correlation correction from the next-to-valence shell is not that unimportant. According to the MBPT(2) estimates of this contribution, correlating the M shell in HBr and N shell in HI increases the corresponding dipole moments by about 0.005 a.u. The effect of the higher-order contributions from those shells is expected to be rather small.

The difference between the final equilibrium value of the dipole moment of HCl (0.421 a.u.) and the corresponding SCF result (0.467 a.u.) is, in principle, predicted at the level of nonrelativistic approach. However, for HI the relativistic and correlation contributions to μ are almost equal to each other. Even the intermediate case of HBr already shows that both these contributions should be taken into account.

The analysis of the electron correlation contribution to the calculated dipole moments shows that the present perturbation scheme for estimating the relativistic corrections to μ works quite satisfactory. Including this correction and simultaneously accounting for the incomplete adequacy of the MBPT(4) treatment of the electron correlation contribution brings the theoretical values of μ for HBr and HI into good agreement with the experimental data.

The relativistic multireference SD-CI method[14] has recently been employed for the calculation of dipole moments of HBr[9] and HI[7] It is rather difficult to carry out a direct comparison between those and the present results. Chapman *et al.*[7,9] employ a rather advanced method for the treatment of relativistic effects. On the other hand, (i) they use the effective core potentials, and thus neglect the core contributions to μ at both the SCF and CI levels, (ii) they assume that the Hellman–Feynman theorem is satisfied for MRSD-CI wave functions. The accuracy of the effective core potential approximation in relativistic CI calculations of molecular electric properties has yet to be verified. Rather spectacular results of recent relativistic calculations of molecular dipole moments[7,9] do not account for the possible modification of the effective core potential in the presence of the external electric field. As regards the second assumption of Chapman *et al.*,[7,9] the difference between the expectation value and energy derivative data for μ can be also important.[23] Moreover, the limited CI approach suffers from the presence of unlinked cluster contributions to μ.[51] In spite of using the expectation value definition of μ and neglecting the unlinked cluster contributions, Chapman *et al.*,[7,9] obtained a very good agreement between the calculated and experimental dipole moments. This suggests that the abovementioned deficiencies of their approach are either irrelevant or lead to certain mutual cancellation of different errors. A finite-field perturbation evaluation of μ within the relativistic CI approach[3,7,9] followed by the estimation of the unlinked cluster terms[51] may easily resolve this problem.

It should be clearly stated that the present perturbation approach based on the truncated MBPT series for the correlation contribution and a simple estimate of the relativistic correction to molecular SCF dipole moments is not expected to give the exact values of μ. It is quite sufficient that the method gives a reliable value of the relativistic contribution to the dipole moment in the low-order approximation.

The vibrational contributions[41] to the dipole moment of hydrogen halides are almost negligible as compared with other corrections to the SCF values. Moreover, the vibrational corrections to μ evaluated for hydrogen halides are relatively independent of the method used to compute the total molecular energies.

The relativistic effects on the dipole polarizability of hydrogen halides appear to be small. The relativistic contribution to $\alpha_{\parallel}$ is practically zero for HCl and HBr. In the case of HI, it leads to a small reduction of the total polarizability. However, the generalization of these conclusions as allowed only under the condition of negligibly small spin–orbit interaction effects which are not accounted for by the present approximation for H^{01}.

The results for dipole moments and polarizabilities of hydrogen halides are based on the assumption of the validity of Eq. (18). The approximate separability of the relativistic and electron correlation contributions has a rather sound basis in the case of those properties; the regions of the density distribution, which are mostly affected, are well separated. However, in the case of large relativistic effects, the interference between the two perturbations might be of some importance.

As discussed in Sec. II the methods which globally account for the electron correlation and are based on a multiconfiguration form of the approximate wave function can lead to estimates of the total first-order relativistic contribution to $Q^{(k)}$, i.e.,

$$Q^{(k,1)} = \sum_{m} Q^{(k,1,n)}. \qquad (24)$$

To this class of methods belongs the MRSD-CI approach[14] employed by Chapman *et al.*[7,9] The inconvenience of their method is that the Hellman–Feynman theorem does not hold for a MRSD-CI wave function. From this point of view the MCSCF methods are given some preference. On the other hand, the amount of the electron correlation effects recovered by these methods is usually small. Thus, the estimate of the first term in Eq. (13) is expected to be rather poor.

As long as the given system is reasonably well described in the single-determinant SCFHF approximation a useful estimate of contribution (24) can be obtained from simple MCSCF functions. This is illustrated by the CASSCF data presented in Sec. III C.

The calculations for the AgH molecule have been carried out at a single internuclear distance of 3.30 bohr with the main purpose to estimate the magnitude of the relativistic contribution to the electric dipole moment and to study the interplay between the relativistic and electron correlation effects. Our SCF energy data presented in Table VI should be virtually the same as those of Martin[5] who employed the Cowan–Griffin operator in first-order perturbation calculations of relativistic corrections to spectroscopic constants of the $X\,^1\Sigma^+$ state of AgH. The present first-order estimate of the relativistic contribution to the SCF energy is reasonably close to the variation result ($-113.410\,824$ hartree) obtained recently by Hess and Chandra.[6]

The dipole moment data show that for AgH the relativistic effects are as important as the electron correlation contribution. The difference between the CASSCF values of the first-order relativistic contribution to μ and the $Q^{(1,1,0)}$ value obtained from SCF calculations gives a measure of the interference between the valence correlation and relativistic effects. For the $(4\sigma 2\pi 2\delta)$ active space and 12 correlated electrons, this difference amounts to about 10% of the total first-order relativistic contribution, i.e., to about 0.02 a.u. Such a contribution should not be neglected in accurate calculations of molecular dipole moments.

The dipole polarizability of AgH experiences a rather negligible influence of the relativistic perturbation. The electron correlation effects play in this case the dominant role. Such behavior of $\alpha_{\parallel}$ could have been guessed on the basis of the electronic structure of AgH. Most of the dipole polarizability of AgH comes from the polarizability of the diffuse charge distribution at the negatively charged hydrogen. Hence, the electron correlation effects on α should follow those for the H^- ion.[52] The very large electron correlation contribution[52] to the dipole polarizability of H^- is to some extent quenched by the presence of the positively charged Ag atom. Even so, the correlation effect on α of AgH is still quite large. On the other hand, the relativistic perturbation at the H atom in AgH is virtually negligible.

There seem to be no experimental data available for either the dipole moment or polarizability of AgH. However, a very useful reference for the present dipole moment data follows from a series of accurate nonrelativistic calculations by McLean.[53] The SCF dipole moment at $R = 3.30$ bohr interpolated from his data (-1.992 a.u., Table IV in Ref. 53) agrees well with our nonrelativistic value of -2.0334 a.u. On the basis of a series of MCSCF and SD-CI calculations McLean concludes that the nonrelativistic dipole moment of AgH should be -1.81 ± 0.08 a.u. The nonrelativistic CASSCF result of this paper are within the error bars estimated by McLean.

Another important study of the dipole moment of AgH has been recently carried out by Langhoff *et al.*[54] A comparison of their nonrelativistic SD-CI and CPF data indicates a considerable contribution of unlinked clusters to the SD-CI dipole moment and one may wonder how large this contribution could be for the relativistic multireference SD-CI method.[7,9] Some idea about the importance of the relativistic contribution to the dipole moment of AgH can be gained by comparing the nonrelativistic CPF data of Langhoff *et al.*[54] and their CPF results calculated with relativistic effective core pseudopotentials (Table XII of Ref. 54, CPF RECP 12-el results). The corresponding difference amounts to about 0.164 au.

The most recent calculation by Balasubramanian[55] which has been performed at the level of the relativistic MRSD-CI approximation with effective core pseudopotentials gives the dipole moment value of -1.20 a.u. at $R = 3.092$ bohr. This result is almost the same as the MCPF value obtained by Langhoff *et al.*[54] in a nonrelativistic approach with relativistic core pseudopotentials (-1.156 a.u.).

The present largest CASSCF calculation with relativistic corrections estimated through the first-order in H^{01} of Eq. (17) predicts the dipole moment of about -1.54 a.u. at $R = 3.30$ bohr. The parallel component of the dipole polarizability tensor of AgH is predicted to be of order of 50 a.u. Both these results, however, may need further refinement. The basis set[5,6] used in the present study is not expected to be highly efficient in calculations of electric properties of AgH. Its choice[5,6] has been made rather for the purpose of an adequate representation of the relativistic perturbation. Moreover, the electron correlation contribution may not be sufficiently accounted for by a rather small active space chosen to describe as many as 12 electrons.[54,55] Thus, the present data for AgH should be rather considered as an illustration of the main principles exposed in Sec. II.

V. SUMMARY AND CONCLUSIONS

We have presented a simple method for the estimation of the relativistic contribution to molecular properties from nonrelativistic wave functions. In the absence of important spin–orbit interactions the relativistic molecular Hamiltonian is assumed to be well enough approximated by the usual nonrelativistic term and the Cowan–Griffin operator. With this assumption a convenient form of the perturbation approach is developed which takes into account the external perturbation leading to the property of interest and the internal (formal) perturbations which arise from the electron correlation and relativistic effects.

The theory described in this paper has been applied to the calculation of molecular electric properties. The dipole moments and polarizabilities of hydrogen halides have been calculated by using the finite-field MBPT techniques. The present estimates of the relativistic contribution to the dipole moment of HCl, HBr, and HI show its increasing role in accurate predictions of electric properties of those molecules. The HCl dipole moment can be reliably calculated in the nonrelativistic approximation. The relativistic correction becomes already non-negligible for HBr and must be certainly accounted for in any sophisticated evaluation of the dipole moment of HI.

The relativistic effects on the dipole polarizability of hydrogen halides turn out to be rather small. This follows from the fact that the outer regions of the electronic density distribution, which are of primary importance for the dipole polarizability evaluation, are only insignificantly affected by the first-order relativistic terms.

We have also studied the role of the core correlation contribution to electric properties of the hydrogen halide molecules. It has been found that the electron correlation effect of the next-to-valence shell is not as small as usually assumed in the majority of calculations for molecules with heavy atoms.

The interplay between the relativistic and correlation contributions to molecular electric properties has been studied for the ground electronic state of the AgH molecule. It has been found that 90% of the total calculated relativistic contribution to the dipole moment of AgH is recovered at the SCF level of approximation. It appears that the interference between the electron correlation and relativistic effects is not highly important in the absence of significant spin–orbit interactions.

The method presented in this paper is meant only to give some realistic estimates of the relativistic contribution to molecular properties. This, however, might be sufficient for resolving several discrepancies between accurate nonrelativistic and experimental data.[8,20,21] In the finite-field perturbation framework the present method offers a convenient computational tool for the evaluation of the relativistic contribution to a variety of molecular properties.

The illustrative material presented in this paper corresponds to the case when the external (electric field) and relativistic perturbations operate in almost disjoint regions of the electron density distribution. This is certainly not the case for the external perturbation resulting in the electric field gradient. In this case the relativistic effect on the calculated electric field gradient easily becomes the major factor which determines the accuracy of the calculated molecular property. The corresponding investigations are in progress and will be published elsewhere.[57]

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